

PRADEEP KUMAR

AI ENGINEER · AGENTIC SYSTEMS · RETRIEVAL · LLM PLATFORMS

Bengaluru, India | +91 9980012759 | pradeepkumar94p@gmail.com | linkedin.com/in/pradeepk21 | github.com/pradeepA2125

SUMMARY

AI Engineer with 6 years shipping production LLM and agentic systems at client scale. Lead Engineer at IBM — enterprise RAG serving 5,000+ users, multi-agent workflows on watsonx Orchestrate, and USD 534K influenced revenue in H1 2026. Author of **Crucible**, a 120k-LOC open-source coding agent (Python, TypeScript, Rust) shipped as a VS Code extension, with an LSP-resolved code graph, hybrid-recall memory layer, MCP client, and 9 integrated model providers.

OPEN SOURCE

Crucible — Agentic coding assistant for VS Code

2026

github.com/pradeepA2125/crucible | Python, TypeScript, Rust | 120k LOC | Apache 2.0

- FastAPI orchestrator running ReAct tool loops against isolated shadow workspaces with transactional promote/rollback, so no partial edit ever reaches the user's repository.
- Rust incremental indexer producing an LSP-resolved symbol graph (tree-sitter plus pyright, typescript-language-server, rust-analyzer, gopls, jdtls) across 6 languages, giving the agent call and inheritance edges rather than plain text search.
- Cross-session memory layer — context compaction with anchored summarization, hybrid recall over sqlite-vec and BM25, and cross-encoder reranking.
- 9 LLM providers behind one interface with runtime capability probing and graceful degradation; MCP client with per-tool approval gates.
- Shipped as an installable VS Code extension that provisions and supervises its own runtime, with a checksummed cross-platform release pipeline on GitHub Actions.

EXPERIENCE

Lead Engineer
IBM

Feb 2024 – Present
Bengaluru, India

- Architected and deployed 3+ enterprise-grade RAG systems on watsonx.ai and watsonx.orchestrate across finance, HR, and energy domains, serving **5,000+** internal and client users; built the supporting inference, ranking, and feedback microservices, containerized with Podman on IBM Cloud Code Engine with auto-scaling, absorbing 3x traffic growth without redesign.
- Promoted from Senior AI Engineer to **Lead Engineer**; served as Technical Lead for the Rogers data-intelligence engagement, delivering the Phase 1 deployment of Cloud Pak for Data and IBM Knowledge Catalog (now watsonx.data intelligence) onto customer-managed OpenShift via oc and cpd-cli — authored the high-availability cluster sizing and hardware specification and resolved recurring operator reconciliation failures to reach a stable production install.
- Designed end-to-end retrieval pipelines (ingestion, chunking, embedding, vector indexing, ranking, LLM synthesis) processing **1M+ documents at sub-2s** median latency; implemented hybrid search across Elasticsearch and Watson Discovery, improving retrieval precision by 28% and reducing hallucination-related escalations by 35%.
- Designed structured JSON contract-based outputs and observability layers for tool-calling pipelines, reducing parsing failures by **90%** and debugging time by 45%; drove evaluation-driven prompt engineering, LoRA adapter fine-tuning, and grounding strategies, improving answer consistency by 22% across evaluation benchmarks.
- Built multi-agent orchestration workflows for resume-to-JD matching, document analysis, and compliance-driven generation, cutting manual review time by 40%; extended the agent portfolio on watsonx Orchestrate with a LangGraph natural-language-to-SQL tool, a vision-language document extraction agent, and a call-transcript analyzer producing structured insight across six dimensions.
- Built **Ask-Git**, a conversational code-intelligence assistant over Git history — LangChain, GitHub APIs, and vector stores supporting natural-language querying of commit history, file-level changes, and pull-request metadata across multi-repository configurations; adopted into IBM's CE Coach enablement accelerators and reused across multiple client engagements.
- Influenced **USD 534K** in customer revenue (USD 200K annual recurring) across three strategic engagements, supporting the Revenu Québec BAMOE opportunity from delivery through closure.

- Drove watsonx Orchestrate adoption across APAC — platform training for regional teams, ADK upskilling, bootcamp content, and the "Getting Started with watsonx Orchestrate" QuickStart guide; published an OpenShift deployment knowledge base selected as a formal IBM reusable asset; mentored engineers and led architecture reviews for enterprise AI platform initiatives.

Senior AI Engineer

Sep 2021 – Feb 2024

Infilect

Bengaluru, India

- Led 6 end-to-end AI deployments across retail analytics for global FMCG clients.
- Built scalable computer vision systems for Share-of-Shelf and On-Shelf Availability analytics.
- Developed image retrieval and segmentation pipelines (Patch2Pix, SAM), improving inference efficiency by 25%, and DINO-based retrieval that reduced manual workflows by 35%.
- Built cross-modal LLM-based services that cut manual annotation effort by 40%.
- Architected deep learning microservices using Docker, Kubernetes, Redis, nginx, and TensorRT.

Computer Vision Engineer

Oct 2020 – Sep 2021

Drishte

Bengaluru, India

- Built tracking, detection, classification, and re-identification models for large-scale crowd analytics systems.
- Developed Upasthiti, a touchless face recognition attendance system optimized for mobile devices.
- Deployed models to Jetson Nano and edge devices with TensorRT, OpenVINO, and TensorFlow Lite; improved accuracy by 20% and automated CI/CD to cut deployment time by 30%.

SKILLS

Agentic & LLM: LangGraph, LangChain, CrewAI, Model Context Protocol (MCP), ReAct tool loops, multi-agent orchestration, function/tool calling, structured outputs and JSON-schema constrained decoding, context compaction, evaluation-driven prompt engineering, LoRA fine-tuning, vision-language models (VLMs)

Retrieval: RAG, hybrid search, Elasticsearch, FAISS (IVF, PQ), sqlite-vec, BM25, cross-encoder reranking, chunking and embedding pipelines, natural-language-to-SQL

Model Providers: OpenAI, Anthropic, Gemini, Groq, watsonx.ai, vLLM, Ollama, OpenRouter, NVIDIA NIM

Languages: Python, TypeScript, Rust, C++, Java, SQL

Platform & Infra: OpenShift (oc, cpd-cli), Kubernetes, Docker, Podman, FastAPI, Kafka, Redis, TimescaleDB, PostgreSQL, Terraform, Vault, AWS, Azure, GCP, GitHub Actions, on-premise and restricted-network deployment

IBM Platform: watsonx.orchestrate (ADK), watsonx.ai, watsonx.governance, IBM Knowledge Catalog (watsonx.data intelligence), Cloud Pak for Data, Watson Discovery, DataStage

Deep Learning: PyTorch, TensorFlow, OpenCV, TensorRT, OpenVINO, SAM, BLIP-2, DINO

EDUCATION

BE, Electronics and Communication

SLN College, VTU, Karnataka

2016 – 2020

CGPA: 8.0/10

CERTIFICATIONS

O'Reilly AI Agents · NVIDIA Fundamentals of Accelerated Computing (CUDA C/C++) · HashiCorp Terraform and Vault

IBM: watsonx Orchestrate, IBM Knowledge Catalog, DataStage, Data Lineage (Manta), Concert, watsonx BI · Udacity

Nanodegrees: Machine Learning, Deep Learning, Computer Vision, NLP, Deep Reinforcement Learning